

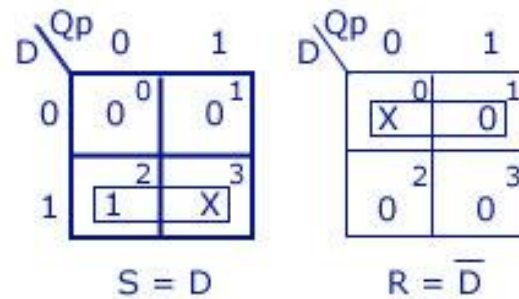
FlipFlop name	Characteristic Table			Characteristic Equation	Excitation Table			
SR	S	R	Q_{n+1}	$Q_{n+1}=S+R' Q_n$	Q_n	Q_{n+1}	S	R
	0	0	Q_n		0	0	0	X
	0	1	0		0	1	1	0
	1	0	1		1	0	0	1
	1	1	invalid	Where SR=0	1	1	X	0
JK	J	K	Q_{n+1}	$Q_{n+1}=J Q_n'+K' Q_n$	Q_n	Q_{n+1}	J	K
	0	0	Q_n		0	0	0	X
	0	1	0		0	1	1	X
	1	0	1		1	0	X	1
	1	1	Q_n'		1	1	X	0
D	D		Q_{n+1}	$Q_{n+1}=D$	Q_n	Q_{n+1}	D	
	0		0		0	0	0	
	1		1		0	1	1	
					1	0	0	
					1	1	1	
T	T		Q_{n+1}	$Q_{n+1}=T Q_n'+T' Q_n$	Q_n	Q_{n+1}	T	
	0		Q_n		0	0	0	
	1		Q_n'		0	1	1	
					1	0	1	
					1	1	0	

S-R Flip Flop to D Flip Flop

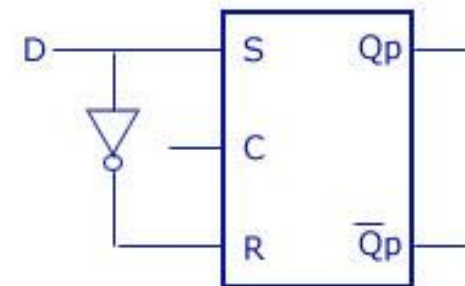
Conversion Table

D Input	Outputs		S-R Inputs	
	Q_p	Q_{p+1}	S	R
0	0	0	0	X
0	1	0	0	1
1	0	1	1	0
1	1	1	X	0

K-maps



Logic Diagram

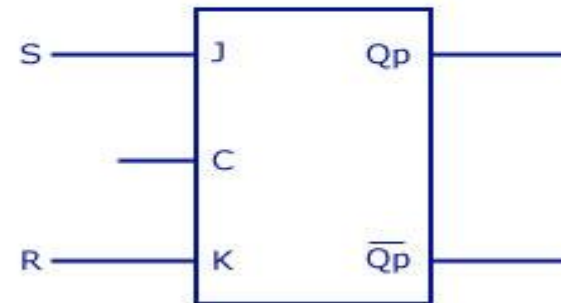


J-K Flip Flop to S-R Flip Flop

Conversion Table

S-R Inputs		Outputs		J-K Inputs	
S	R	Q _p	Q _{p+1}	J	K
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	X	1
1	0	0	1	1	X
1	0	1	1	X	0
1	1	Invalid		Dont care	
1	1	Invalid		Dont care	

Logic Diagram



S	RQ _p			
	00	01	11	10
0	0 ⁰	X ¹	X ³	0 ²
1	X ⁴	X ⁵	X ⁷	X ⁶

$$J=S$$

K-maps

S	RQ _p			
	00	01	11	10
0	X ⁰	0 ¹	1 ³	X ²
1	X ⁴	0 ⁵	X ⁷	X ⁶

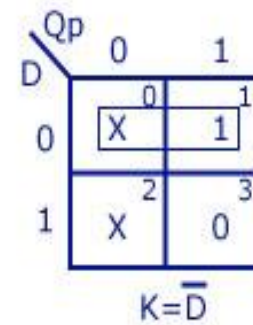
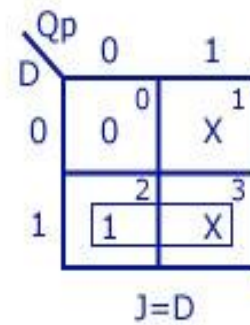
$$K=R$$

J-K Flip Flop to D Flip Flop

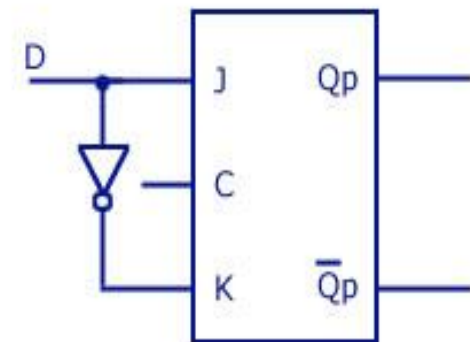
Conversion Table

D Input	Outputs		J-K Inputs	
	Q_p	Q_{p+1}	J	K
0	0	0	0	X
0	1	0	X	1
1	0	1	1	X
1	1	0	X	0

K-maps



Logic Diagram

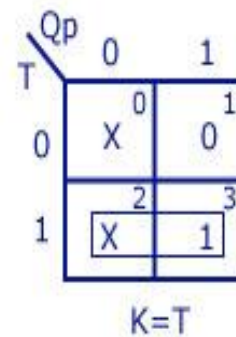
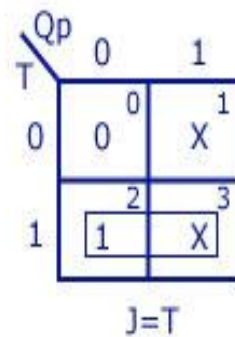


J-K Flip Flop to T Flip Flop

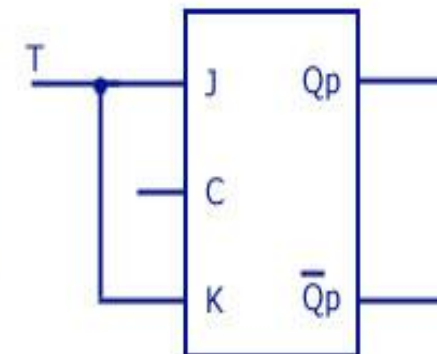
Conversion Table

T Input	Outputs		J-K Inputs	
	Q_p	Q_{p+1}	J	K
0	0	0	0	X
0	1	1	X	0
1	0	1	1	X
1	1	0	X	1

K-maps



Logic Diagram



D Flip Flop to J-K Flip Flop

Conversion Table

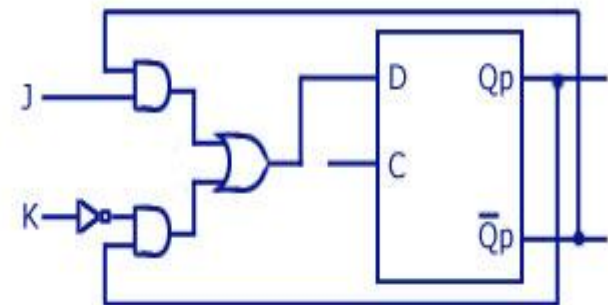
J-K Input		Outputs		D Input
J	K	Q _p	Q _{p+1}	
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

K-map

KQ _p		00	01	11	10
J	0	0	1	0	0
1	1	1	1	0	1

$$D = J\bar{Q}_p + \bar{K}Q_p$$

Logic Diagram



D Flip Flop to S-R Flip Flop

Conversion Table

S-R Inputs		Outputs		D Input
S	R	Q _p	Q _{p+1}	
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	Invalid		Dont care
1	1	Invalid		Dont care

K-map

S	RQ _p	00	01	11	10
		0	1	3	2
0	0	0	1	0	0
1	1	4	5	7	6
		1	1	X	X

$$D = S + \bar{R}Q_n$$

Logic Diagram

